

“ I... FOUND A PORTION OF THE LUNG AS LARGE AS A TURKEY’S EGG, PROTRUDING THROUGH THE EXTERNAL WOUND, LACERATED AND BURNT. ”

William Beaumont, US army surgeon, from *Experiments and Observations on the Gastric Juice, and the Physiology of Digestion*, 1833



US army surgeon William Beaumont inserted a tube into the stomach of his colleague Alexis St. Martin, who had been injured by a gunshot.

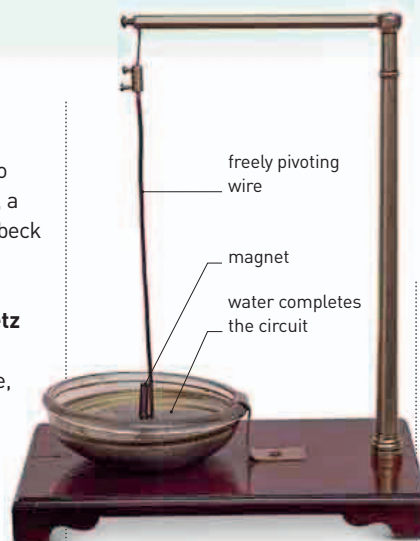
BUILDING ON DANISH PHYSICIST HANS CHRISTIAN ØRSTED’S discovery in 1820 that an electric current makes a magnet move, British scientist **Michael Faraday** (see 1837) showed that a wire carrying an electric current moved in a circle around a fixed magnet, and that a suspended magnet moved in a circle around a fixed wire carrying a current. He had discovered **electromagnetic rotation**—the principle of the electric motor.

German-Estonian scientist **Thomas Seebeck** (1770–1831) observed that a compass needle wavers when it is close to a loop of two different metals that are cooled in one place and heated in another. This is because the

slightly different movement of heat through each of the two metals disturbs atoms to generate an electric current, a phenomenon called the Seebeck or **thermoelectric effect**.

In the field of geology, Swiss geologist **Ignatz Venetz** (1788–1859) suggested that in the past, during an ice age, the world was colder and Europe was covered in glaciers that shaped much of its landscape.

On the English coast, near Lyme Regis in Dorset, fossil collector **Mary Anning** (see 1810–11) found the first fossil *plesiosaur*—a huge marine reptile that lived 65–195 million years ago. The following year, another fossil hunter, **Gideon**



Mantell (1790–1852) found teeth of a huge reptile he called an *iguanodon*, which was later identified as a dinosaur. In Yorkshire, British naturalist **William Buckland** (1784–1856)

Faraday's experiment

A replica of the apparatus Michael Faraday used to demonstrate the principle of electromagnetic rotation, which is the basis of electric motors.

discovered ancient remains of a hyena's den, with bones of rhinoceroses, elephants, and lions, showing that wildlife in the British Isles was once very different.

Geologists were starting to identify the different ages in Earth's past from fossils found in rocks. In 1822, British geologists **William Phillips** (1775–1828) and **William Conybeare** (1787–1857) made the **first identification of a geological period**. They named it the **Carboniferous period** after

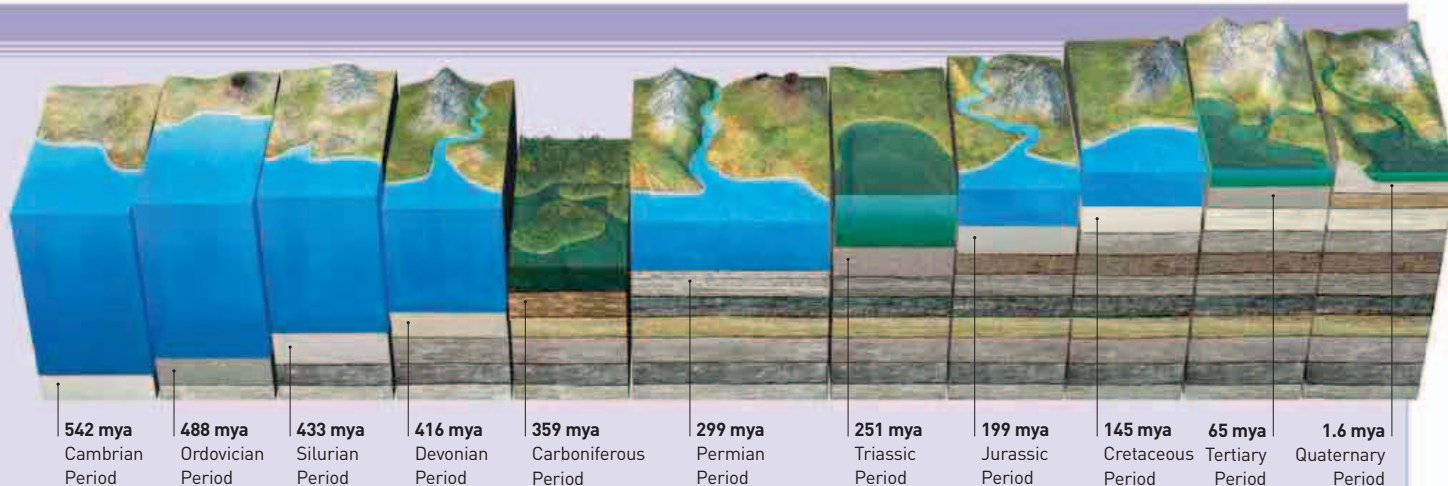
the carboniferous (coal-bearing) strata of northern England.

The same year, British computer pioneer **Charles Babbage** (1791–1871) proposed his ingenious idea for a **Difference Engine**: a calculating machine built from cogs and rods that would work automatically and eliminate human error.

American army surgeon **William Beaumont** (1785–1853) was the first person to observe human digestion in the stomach. He performed experiments on a soldier who had been shot in the abdomen, and pioneered **gastric endoscopy**—inserting a tube to look inside the stomach.

GEOLOGICAL PERIODS

Layers of rock form one on top of the other, with the oldest at the bottom, unless they have been disturbed. The sequence forms the geological column and is the basis for dividing Earth's history into geological periods, identified originally by the fossils found in each rock layer. The oldest, deepest layer is the Cambrian Period 542–488 million years ago—the first era when life left enough fossils to date it.



1821 Ignatz Venetz proposes his ice age theory

July 3, 1822 Charles Babbage publishes proposal for his difference engine

1821 Thomas Johann Seebeck discovers the thermoelectric effect

1821 Mary Anning finds the first plesiosaur fossil

1821 French physicist Augustin-Jean Fresnel completes his wave theory on the polarization of light

1821 British scientist Michael Faraday discovers electromagnetic rotation

1822 Gideon Mantell discovers fossil teeth of an iguanodon

1822 Belgian geologist Jean-Baptiste-Julien d'Omalus d'Halloy identifies the Cretaceous Period

1822 William Beaumont pioneers human gastric endoscopy